

```
def draw(self):
    pygame.draw.circle(self.window, self.color, (self.centerX, self.centerY),
                        self.radius, 0)
```

Listing 10-8: The `circle` class that inherits from the `Shape` class

The `circle` class also contains the `clickedInside()`, `getArea()`, and `draw()` methods, whose implementation is specific to a circle.

Finally, [Listing 10-9](#) shows the code of the `Triangle` class.

File: InheritedShapes/Triangle.py

```
# Triangle class

import pygame
from Shape import *

class Triangle(Shape):

    def __init__(self, window, maxWidth, maxHeight):
        super().__init__(window, 'Triangle', maxWidth, maxHeight)
        self.width = random.randrange(10, 100)
        self.height = random.randrange(10, 100)
        self.triangleSlope = -1 * (self.height / self.width)
        self.rect = pygame.Rect(self.x, self.y, self.width, self.height)

    def clickedInside(self, mousePoint):
        inRect = self.rect.collidepoint(mousePoint)
        if not inRect:
            return False

        # Do some math to see if the point is inside the triangle
        xOffset = mousePoint[0] - self.x
        yOffset = mousePoint[1] - self.y
        if xOffset == 0:
            return True

        pointSlopeFromYIntercept = (yOffset - self.height) / xOffset # rise over
run
        if pointSlopeFromYIntercept < 1:
            return True
```